

Objectives

Primary biliary cirrhosis (PBC) is a chronic liver disease commonly studied using survival analysis. The PBC datasets in the R survival package provide both baseline measurements and longitudinal follow up data for clinical biomarkers.

The goal of this project is to compare how different survival models influence estimated covariate effects in PBC patients:

- Baseline Cox proportional hazards model
- Extended Cox model with time varying covariates
- Fine–Gray competing risks model treating transplantation as a competing event

Data and Exploratory Analysis

Data Sources

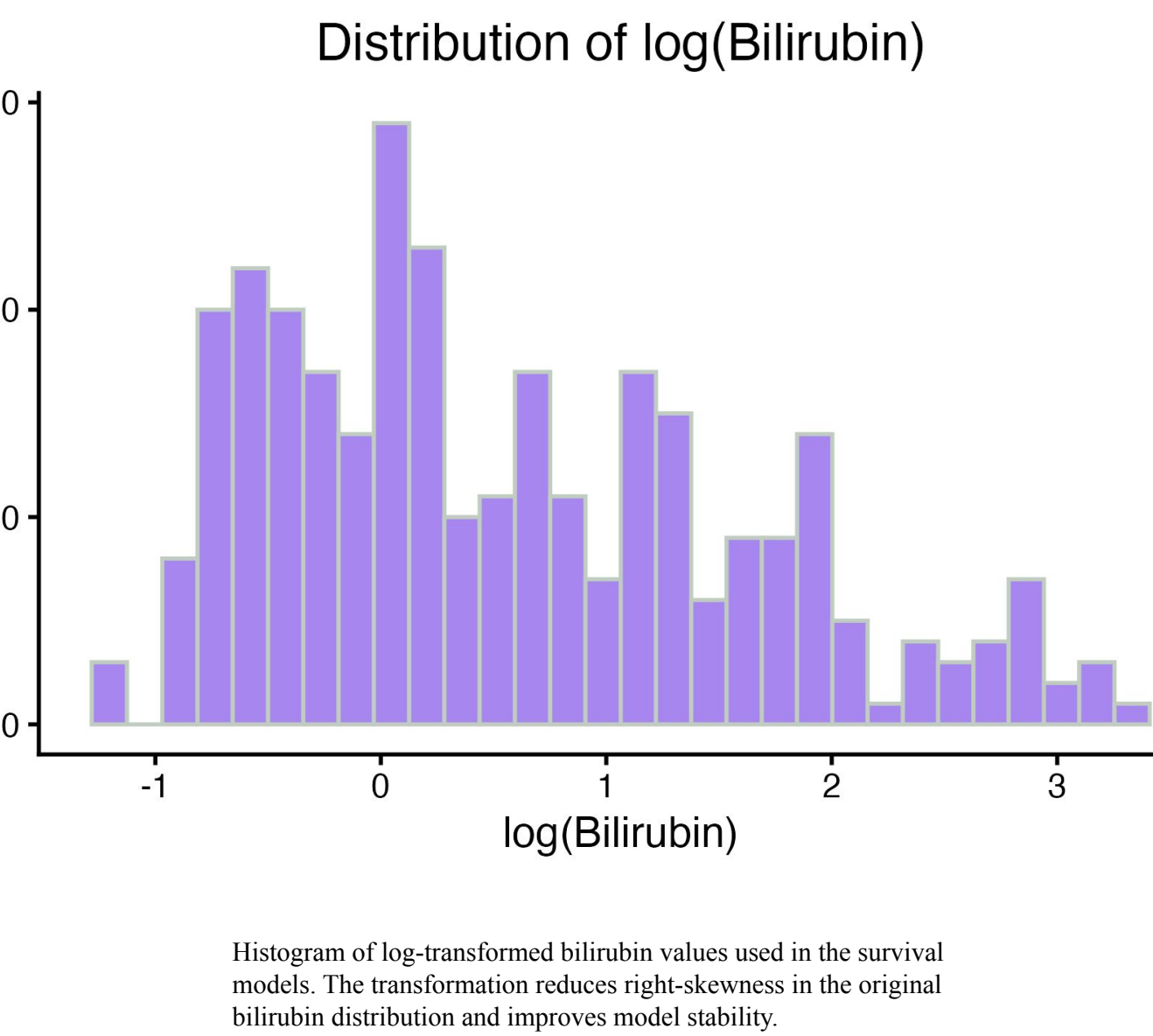
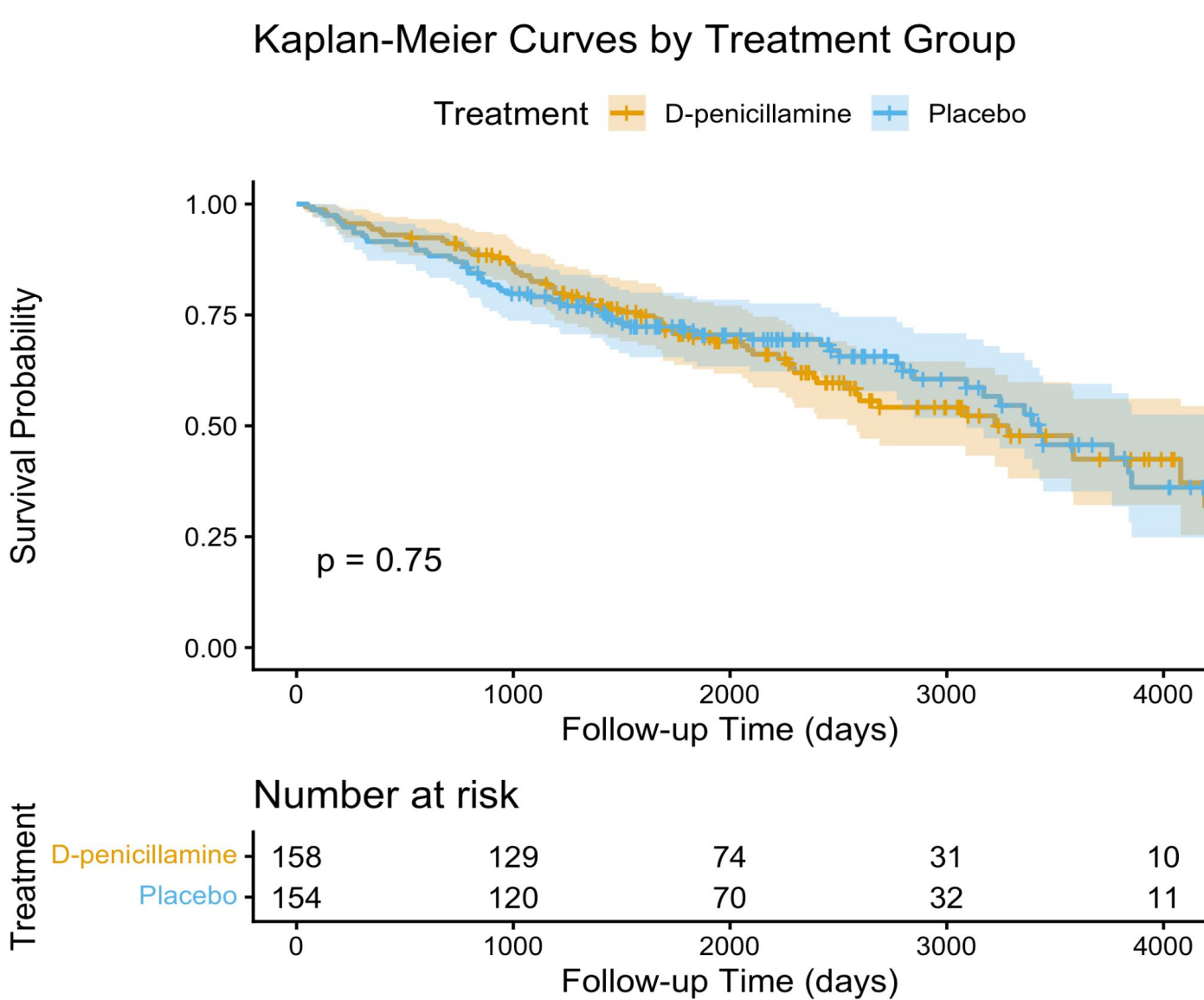
Two datasets from the survival package in R were used: pbc and pbcseq. The pbc dataset contains baseline clinical information and follow-up outcomes for patients with primary biliary cirrhosis, including death, transplantation, and censoring. The pbcseq dataset contains repeated laboratory measurements collected over follow-up for the randomized trial patients, which allows the incorporation of time-varying covariates in an extended Cox model.

Data Processing

For the baseline analysis, the pbc dataset was cleaned and reorganized to retain patient identifier, follow-up time, event status, demographic variables, clinical indicators, laboratory measurements, and histologic stage. Death was defined as status = 2, transplantation as status = 1, and censoring as status = 0. Patients with missing treatment assignment were excluded to maintain consistency with the randomized study population.

Exploratory analysis showed that bilirubin was strongly right-skewed. To reduce skewness and improve model stability, a logarithmic transformation was applied, producing the variable log_bili. Univariate Cox models were first fitted for candidate covariates, and variables showing stronger evidence of association with mortality were considered in the multivariable analysis. The final baseline covariates retained for model comparison were age, sex, edema, log-transformed bilirubin, albumin, prothrombin time, and histologic stage.

For the time-varying analysis, the pbcseq dataset was converted into counting process form. For each patient, consecutive visit times were used to define start and stop times for risk intervals. Specifically, tstart was set to the current visit day and tstop to the next visit day, with the final interval ending at the observed follow-up time. A death indicator was assigned only to the final interval when death occurred. This ensured that each patient contributed at most one death event while allowing covariate values to vary across follow-up intervals.



Kaplan–Meier Survival Curves

Kaplan–Meier curves were used to explore the overall survival pattern in the PBC dataset because they provide a nonparametric estimate of survival before regression modeling. Survival probabilities over follow-up appear similar between the D-penicillamine and placebo groups, and the log-rank test does not indicate a significant difference (p = 0.75). This exploratory analysis provides a descriptive overview of survival trends prior to formal regression modeling.

Statistical Models

Three survival models were fitted to examine covariate effects on the hazard of death in patients with primary biliary cirrhosis.

Baseline Cox proportional hazards model

A Cox proportional hazards model was fitted using baseline covariates from the pbc dataset.

$$h(t | X) = h_0(t)e^{\beta^T X}$$

where X includes age, sex, edema, log(bilirubin), albumin, protime, and stage. Death was treated as the event of interest and transplantation was treated as a censoring event. Hazard ratios represent multiplicative effects of covariates on the instantaneous hazard of death.

Extended Cox model (time-varying covariates)

To incorporate repeated biomarker measurements during follow-up, an extended Cox model was fitted using the pbcseq dataset in counting process form. Covariates were allowed to vary across follow-up intervals.

$$h(t | X(t)) = h_0(t)e^{\beta^T X(t)}$$

where X(t) includes age, sex, edema(t), log(bilirubin(t)), albumin(t), protime(t), and stage(t).

The model was estimated using the counting process formulation:

$$\text{Surv}(t_{\text{start}}, t_{\text{stop}}, \text{event}) \sim X(t) + \text{cluster}(\text{id})$$

The cluster(id) term provides robust standard errors to account for within-subject correlation caused by repeated observations from the same patient.

Fine–Gray competing risks model

Because liver transplantation precludes the occurrence of death under the original risk process, a competing risks analysis was performed using the Fine–Gray subdistribution hazards model. Death was treated as the event of interest and transplantation as the competing event. The model directly estimates subdistribution hazard ratios that describe the effect of covariates on the cumulative incidence of death in the presence of competing events.

Model comparison

Results from all models were summarized using hazard ratios or subdistribution hazard ratios with 95% confidence intervals. Forest plots were used to compare covariate effects across models. This comparison highlights differences in estimated effect sizes when time-varying covariates and competing risks are taken into account.

Results

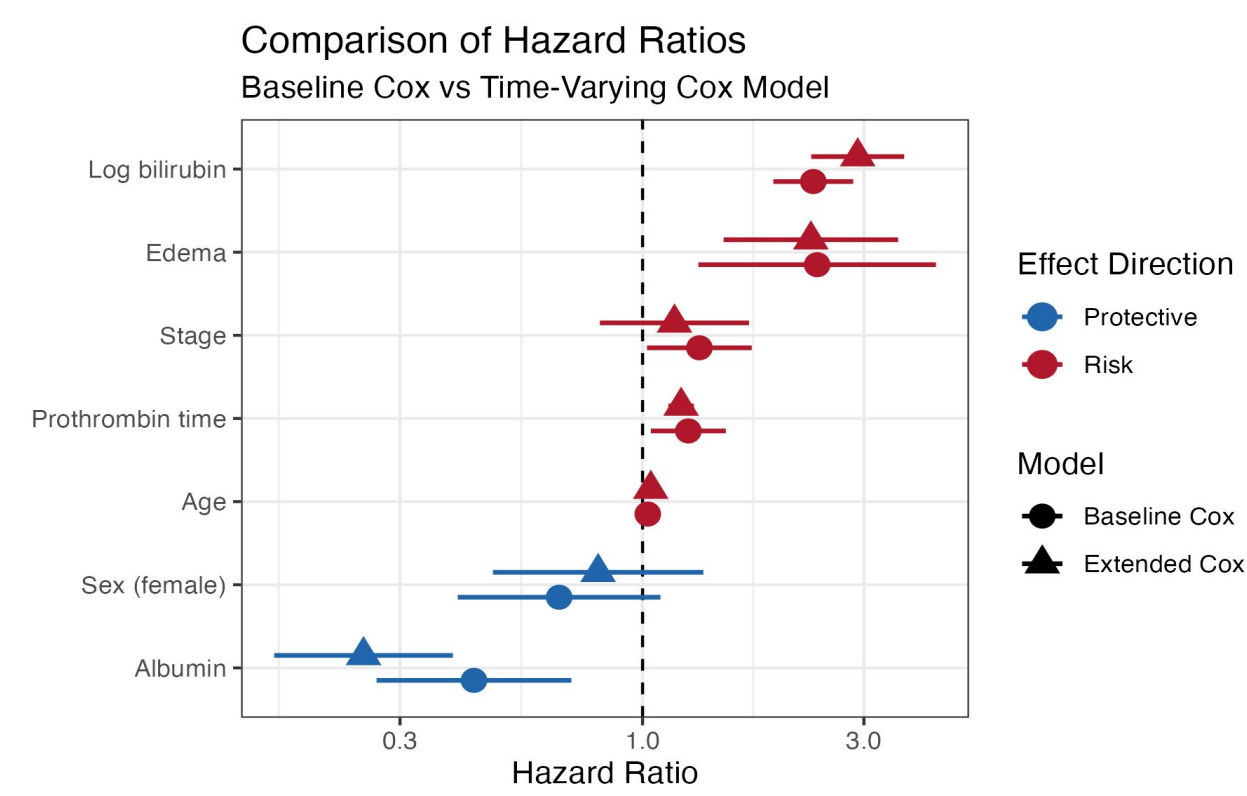
Cox model results and proportional hazards assessment

Baseline Cox model results indicated that higher bilirubin, edema, prothrombin time, and disease stage were associated with increased hazard, while higher albumin showed a protective association. Age had a modest positive association with hazard, whereas sex was not statistically significant. The proportional hazards assumption was assessed using Schoenfeld residual tests. Most covariates satisfied the assumption, although edema (p = 0.032) and prothrombin time (p = 0.014) showed evidence of time-varying effects.

Extended Cox model with time-varying covariates

An extended Cox model using the pbcseq dataset was fitted in counting-process form to incorporate repeated biomarker measurements during follow-up. Covariates were allowed to vary across follow-up intervals and robust standard errors were obtained using clustering by patient.

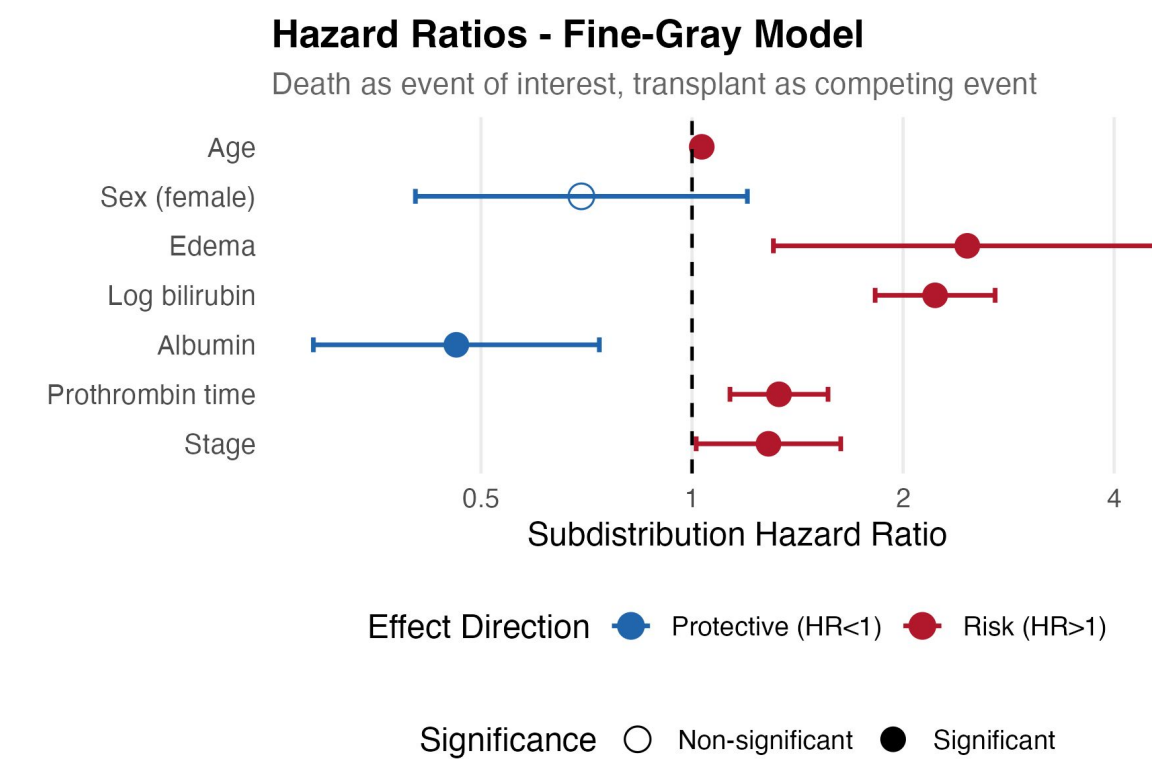
The estimated effects were broadly consistent with the baseline Cox model. Higher time-updated bilirubin remained strongly associated with increased hazard, while albumin showed a protective association. Edema, prothrombin time, and disease stage also maintained positive associations with hazard. This figure compares hazard ratio estimates from the baseline and extended Cox models.



Competing Risks Analysis

Because liver transplantation may occur before death and alters the risk process, a competing risks analysis was conducted using the Fine–Gray subdistribution hazards model.

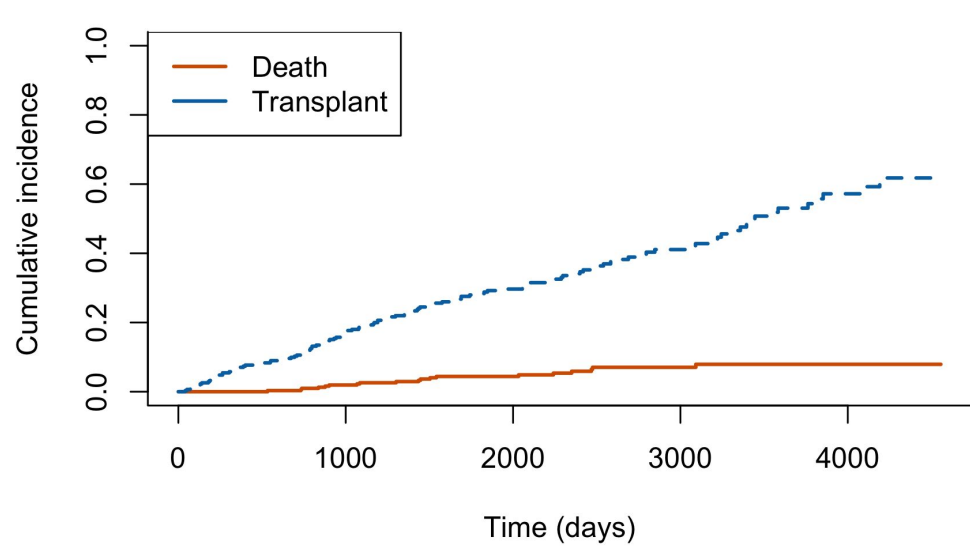
The results were consistent with the Cox models. Higher bilirubin remained strongly associated with increased risk of death (subdistribution HR = 2.22, 95% CI: 1.82–2.71), while higher albumin levels were protective (HR = 0.46, 95% CI: 0.29–0.74). Edema (HR = 2.47) and prothrombin time (HR = 1.33) were also significantly associated with increased risk.



Subdistribution hazard ratios from the Fine–Gray competing risks model (death as event of interest, transplant as competing event). Error bars represent 95% confidence intervals.

Cumulative incidence of events

Cumulative incidence curves illustrate the occurrence of death and liver transplantation over follow-up time. The results show that transplantation occurs more frequently than death and represents an important competing event when estimating the risk of death.



Cumulative incidence curves showing the probability of death and liver transplantation over follow-up time.

Discussion

The results consistently identified several biomarkers associated with the hazard of death in patients with primary biliary cirrhosis. Higher bilirubin, edema, prothrombin time, and disease stage were associated with increased hazard, while higher albumin levels showed a protective association. These findings are consistent across the baseline Cox model, the extended Cox model with time-varying covariates, and the Fine–Gray competing risks model.

Allowing biomarker values to vary over time produced effect estimates similar in direction to those from the baseline Cox model, suggesting that the overall interpretation of covariate effects is robust to the inclusion of longitudinal measurements. Accounting for liver transplantation as a competing event also did not substantially change the interpretation of the main associations.

Overall, these results consistently identify bilirubin, albumin, edema, and prothrombin time as key biomarkers associated with mortality risk in PBC patients.

Limitations

- Some covariates showed evidence of time-varying effects in the proportional hazards test.
- The analysis was based on observational clinical data and may be subject to residual confounding.
- Results depend only on the variables available in the PBC dataset.

References

- T Therneau and P Grambsch (2000), *Modeling Survival Data: Extending the Cox Model*, Springer-Verlag, New York. ISBN: 0-387-98784-3.
- Zhang Z, Reinikainen J, Adeleke KA, Pieterse ME, Groothuis-Oudshoorn CGM. Time-varying covariates and coefficients in Cox regression models. *Ann Transl Med*. 2018 Apr;6(7):121. doi: 10.21037/atm.2018.02.12. PMID: 29955581; PMCID: PMC6015946.